VideoVibes: Enhancing Emotion and Sarcasm Detection in Video Conferencing

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***Abstract*—Online video calls often obscure subtle emotional cues and make sarcasm difficult to detect, leading to miscommunication through this medium. This paper presents VideoVibes, a tool built on top of the Jitsi video-conferencing platform that provides real-time visual cues for emotion and sarcasm during online meetings. VideoVibes uses lightweight facial-expression analysis in the browser to display small emoji indicators on each participant's video tile. A companion backend service processes short audio clips to identify potential sarcasm and briefly displays a crossed-fingers icon when such remarks occur. The system also includes a Learning Mode that gives users short quizzes based on live video to help them practice recognizing emotions independently of the overlays. We describe the design and implementation of the system, the user study conducted to evaluate its usefulness, and the limitations and future directions for expanding both accuracy and accessibility.**

I. Introduction

Remote communication is now a routine part of daily life, but many social cues are lost in video calls. Tone, sarcasm, and emotion can be hard to read through a screen, especially for people who already find it difficult to interpret intent or feeling, such as those on the autism spectrum. Research shows that understanding figurative and sarcastic language depends more on context and language ability than on diagnosis alone [1].

As AI-mediated communication continues to shape how people interact online, it brings both opportunities and ethical challenges. Hancock et al. argue that systems that utilize AI to mediate or augment communication alter the cues people rely on to detect/express empathy, sarcasm, and intent [2]. They emphasize that these tools must be built with transparency and user trust to avoid emotional distortion. This perspective gives VideoVibes a clear ethical foundation to build on.

Studies on emotion recognition and assistive technology suggest that visual feedback can help people better recognize emotions. The Superpower Glass project, for example, used a wearable device to give children with autism real-time emotion cues, leading to better engagement and emotion recognition [3]. Similarly, Dalle Nogare et al. found that emojis can often communicate feelings more clearly than human expressions [4].

Despite these advances, few tools offer real-time emotional or sarcastic feedback during live video calls. VideoVibes addresses this gap by adding small emoji overlays that reflect detected mood and tone. The goal is to make online communication more understandable and inclusive for all users.

II. Background

A. Emotion Recognition in Virtual Communication

There has been a lot of research done into emotion recognition, both using facial expressions and with the use of other technologies such as EEG sensors [5]. These methods show that emotions can be detected through signals like facial movement, tone, and even brain activity, which makes them useful for both in-person and online communication. Building on this foundation, Lin explored how real-time emotion recognition can be directly integrated into video conferencing systems with success and efficiency [6]. Their work supports the feasibility of emotion recognition in virtual environments where subtle cues in the frame can improve social connectedness. The effectiveness of such technologies is proof that the virtual medium does not necessarily block emotional cues, as shown in psychological studies [7].

Although face-to-face interaction differs a lot from relating virtually, emotion recognition is capable of generalizing across these domains [8]. Studies have shown that facial expression recognition systems can still perform effectively in video-based environments, allowing emotional context to be preserved even when physical presence is lost. This holds in spite of people being less facially expressive over video calls than they are in person [9].

Beyond visual cues, combining multiple modalities produces more robust and complete results. Ortega et al. demonstrated that combining audio and video features significantly increased recognition compared to using facial features alone [10]. This work shows how combining multiple signals gives a clearer picture of emotion. Audio signals add temporal context while video features enhance spatial recognition. This provides insight into the VideoVibes system's development, which may expand the feature synchronization.

Several projects have explored ways to use these techniques to support individuals with social or communicative challenges. For example, the Superpower Glass system used a wearable camera and smartphone app to recognize emotions in real time and provide feedback to children with autism, leading to measurable improvements in engagement and emotional understanding [3]. Similar results have been observed in augmented reality learning environments, where interactive visual feedback enhanced social skill development for students with autism [11]. These studies demonstrate that emotion recognition technologies can be effective teaching and assistive tools when used thoughtfully.

We are using face-api.js, a common emotion recognition API that has been used in similar technologies, such as EmojiCam [12]. This API is lightweight, simple, and easy to integrate while being accurate enough for our needs. Its accessibility and real-time processing capabilities make it suitable for VideoVibes, which aims to interpret users' emotions unobtrusively and translate them into clear visual feedback during video calls.

B. Emojis as Visual Aids for Emotion

Emojis provide a simplified, universal way to express emotion in digital communication and can make virtual interactions more intuitive [4]. They act as recognizable visual cues that quickly convey tone, especially when facial expressions or vocal inflection are hard to interpret through a screen.

We are not the first to explore the use of emojis for video-calling applications, but we can learn a lot from those who have. One such technology is EmojiCam, which incorporates emojis into Zoom to simplify virtual interactions [12]. Many of the challenges reported in that study highlight design considerations relevant to VideoVibes. For example, users found that emojis covering participants' entire faces were distracting and sometimes made it harder to recognize emotion. In our implementation, the emoji will appear alongside, rather than overlaying, the participant's face, so both sources of information remain visible. This design lets the emoji guide interpretation while keeping natural facial cues visible.

Research supports the effectiveness of emoji-based feedback for conveying emotion clearly and consistently. Dalle Nogare et al. found that male participants, in particular, identified emotions more accurately from emojis than from real facial expressions [4]. The authors suggested that emojis' simplified and standardized features reduce ambiguity, making them easier to process and recognize. This finding reinforces the potential benefit of using emoji overlays in virtual communication tools, where subtle facial or tonal details can easily be lost due to lighting or camera quality.

Although emojis can clarify meaning, autistic individuals may experience some of the same difficulties with them as with real facial expressions. They have been shown to associate a broader range of emotions with each emoji and to be less accurate than neurotypical participants at recognizing fear, surprise, and sadness in emoji form [13]. Additionally, negative emojis tend to have a stronger influence on message interpretation for these users, which can lead to miscommunication. To reduce this risk, our design allows users to choose which emojis correspond to each emotion and limits the display of incorrect or rapidly changing feedback by averaging emotional responses over time before updating the emoji. Balancing expressiveness and clarity, emoji overlays offer a practical way to make emotions easier to read in video calls.

C. Sarcasm and Nonliteral Language

There is a lot of research showing that autistic individuals often struggle to detect sarcasm and irony in conversation. Sarcastic remarks can appear as often as once every two minutes [1], and understanding them is important for social connection. Studies of figurative language suggest that these difficulties are linked more to language ability than to autism itself, meaning that small differences in verbal reasoning can affect how people interpret nonliteral meaning [14]. Autistic individuals are fully capable of enjoying humor, but their comprehension is highly modality dependent. They often struggle to recognize written or audio-only humor, and they also struggle to identify sarcasm based on tone of voice alone (like in the setting of telecommunication) [15].

This challenge is not limited to people with severe autism. Even autistic individuals who perform well on theory-of-mind (ToM) tests, which measure social reasoning, still have trouble recognizing sarcasm and other nonliteral statements [16]. Panzeri et al. found that performance among autistic children often splits into two groups: some consistently miss ironic intent, while others rely on rule-based reasoning to identify it [17]. This shows that understanding sarcasm can come from explicit reasoning, not just intuition. These findings also suggest that computational tools could use similar cues to help users interpret tone more accurately.

Conveying sarcasm and humor over telecommunication is not a problem exclusive to those with learning disabilities or social disorders. Kruger's social psychology experiment showed all the way back in 2005 that individuals assumed 80% of recipients who received emails they wrote would be able to accurately assess their tone, but only half of recipients correctly assessed the tone of emails they received [18]. This was back before emojis, so perhaps in-text images often tied with emotion have come to close that bias gap. Aforementioned research on emojis supports this claim, and our design with VideoVibes.

The difficulty of detecting sarcasm also extends to technology. In one study comparing four sarcasm-recognition models, the best system reached only 86.32 percent accuracy on the most reliable dataset [19]. Recent reviews of multimodal sarcasm detection report similar challenges, noting that even models combining text, voice, and facial cues still struggle to generalize [20]. Earlier linguistic approaches, such as those using context incongruity between positive and negative phrases, have shown promise but are limited to text-only settings [21]. Together, these studies show that sarcasm detection remains a difficult task, especially in real-time conversations like video calls.

While sarcasm can be hard to recognize, it is a teachable skill. Positive reinforcement, guided feedback, and behavioral methods have all been shown to improve people's ability to detect and respond to sarcastic remarks [1]. A system like VideoVibes could build on these strategies by giving users clear, consistent cues about tone, helping them learn to identify sarcasm on their own, and improving understanding in everyday communication.

D. Assistive and Accessibility Technologies for Social Communication

Many assistive technologies have been designed to help users interpret emotions and social cues more effectively. These tools often aim to make communication easier for individuals who find it challenging to read facial expressions or tone, such as those on the autism spectrum.

One notable example is Superpower Glass, a wearable device that pairs smart glasses with a smartphone app to deliver real-time feedback about others' emotions. In a home-based study with children on the autism spectrum, the system helped users identify emotions more accurately and increased their social engagement [3]. The study demonstrated that lightweight emotion-recognition technology can meaningfully improve everyday communication when integrated into familiar settings.

Recent work has begun extending these ideas directly into video environments. Franzen developed an emotion recognition prototype tailored for autistic individuals in online meetings [22]. Their system used large deep learning models for facial recognition, testing their system in video environments. Results indicated that the detection of emotions such as happiness, sadness, and surprise can be achieved with accuracy and low enough latency for interactive use. It supports the idea that real-time emotion awareness can make communication more inclusive. Overall, emotion recognition in conferencing tools creates a noticeable positive effect on accessibility and social participation.

Other researchers have explored augmented reality systems to achieve similar goals. Sun et al. used interactive AR picturebooks to teach "friendship" and "conflict management" skills to elementary students with autism [11]. After several weeks of lessons, participants showed measurable improvements in social interaction and maintained these gains after the program ended. Although the sample size was small, the results highlight how visual and interactive feedback can support learning and emotional understanding.

Even as recently as this year, researchers have used internet-based multimodal emotion recognition training with positive results in autism spectrum adolescents [23]. The training mimicked everyday social signals instead of just static faces. This live learning environment showed improvements in emotion recognition in its participants, who reported its difficulty to be manageable. VideoVibes takes a similar approach with its quiz mode, focusing on live interaction and ease of use as keys to better emotion recognition.

Beyond emotion recognition, there is a concept of "social presence", defined by Richardson et al. as "a person's sense of being with others in an online environment" [24]. This concept is critical in maximizing the quality of communication through a virtual medium. They emphasize that building social presence relies on emotional feedback and nonverbal cues, both of which VideoVibes helps to decipher. Ultimately, this drives the purpose of VideoVibes, which is to allow users to feel connected and more socially aware during video interactions.

These examples show that technology can play a practical role in supporting social-affective development. VideoVibes extends this idea into virtual communication by providing real-time visual cues during video calls instead of requiring wearable or classroom-based systems. By translating emotional tone into accessible visual feedback, it aims to make digital conversations clearer and more inclusive for a broad range of users.

E. Gaps and Design Implications for VideoVibes

While the literature shows clear potential for emotion-recognition and visual-feedback technologies, several gaps remain between research prototypes and what VideoVibes can achieve within its current scope. Our design builds on prior projects like EmojiCam by positioning emojis beside, rather than over, each participant's face. This preserves visibility of natural expressions while still providing simple emotional cues. However, because of time and resource limits, the system currently uses lightweight open source tools instead of commercial models.

Sarcasm detection presents another challenge. Industry-level solutions often depend on large, multimodal datasets and complex machine learning pipelines that exceed the resources available for a semester-long project. Our heuristic-based approach is designed to demonstrate the concept of tone awareness rather than achieve production-level accuracy. Future iterations could incorporate more advanced sarcasm-detection algorithms or train custom models that account for speech tone and facial expression simultaneously.

User evaluation also remains limited. Ideally, the system would be tested with a diverse group of participants, including individuals on the autism spectrum, to evaluate how effectively it aids emotional understanding. However, time and ethical constraints make such a study unrealistic at this stage. Our prototype instead focuses on usability and concept validation among general users.

Research on awareness cues in video conferencing also offers ideas for future improvements. He et al. introduced LookAtChar, a system that visualizes gaze direction to enhance engagement [25]. It provides insight into the level of clarity which subtle cues can provide to users. VideoVibes likely will not be able to incorporate other features like gaze detection into the model's context and systems overlay output (unless time permits).

Finally, using open source software like Jitsi introduces some technical constraints. While it enables flexible customization, it lacks certain privacy and security features expected in enterprise-grade platforms. Since VideoVibes is a proof of concept, these trade-offs are acceptable for now.

Overall, the current design shows how emotional and tonal feedback can be integrated into real-time communication systems. Future work could focus on expanding sarcasm detection, improving model accuracy, conducting broader user testing, and refining the interface for accessibility and inclusivity.

III. System Design

This section describes the overall structure of VideoVibes and how its major components work together to provide real-time emotion and sarcasm feedback within a video conference.

A. Tech Stack Overview

VideoVibes is built on top of the open-source Jitsi Meet platform, which provides the baseline video-conferencing interface. Our system adds two major capabilities: real-time emotion recognition from webcam video and sarcasm detection from user audio. Emotion processing and visual overlays run entirely in the browser, while audio transcription and sarcasm inference are handled by a lightweight backend.

1) Frontend Architecture: The frontend extends the React-based Jitsi client by adding modules for emotion analysis, sarcasm feedback, and on-screen emoji overlays. Facial expressions are processed locally using `face-api.js`, which periodically analyzes video frames and maps the dominant expression to an emoji. These emojis are drawn as small badges attached to each participant's video tile, allowing natural facial cues to remain visible.

Sarcasm feedback is integrated into the same overlay system. The frontend captures short audio segments, forwards them to the backend, and updates the user's tile whenever a sarcastic utterance is detected. Global Observation and Learning modes are stored in Redux and control whether the interface displays real-time feedback or presents the user with emotion-recognition quizzes. Because all rendering and state management occur in the browser, the system remains lightweight and responsive.

2) Backend Architecture: The backend consists of two small services. A Node.js service streams audio to the Deepgram Speech-to-Text API to obtain transcriptions, and a Python service performs sarcasm classification using a logistic regression model trained on a Reddit sarcasm dataset. The services communicate over simple HTTP requests, and the backend returns a single probability score indicating how likely a given utterance is sarcastic. This modular structure keeps processing costs low and allows the sarcasm model to be updated without modifying the frontend.

3) Infrastructure: VideoVibes uses Jitsi's public infrastructure for conferencing, which handles all media routing, participant layout, and networking. Our additional backend services run locally during development but can be deployed on any lightweight server. Since all emotion analysis happens in the browser and only short audio segments are sent to the backend, the system avoids heavy computation and scales easily for prototype use.

B. User Interface

The VideoVibes interface extends the standard Jitsi Meet layout, preserving its familiar grid of participant video tiles and toolbar controls. Our design adds two elements to this interface: (1) a toggle for switching between Observation Mode and Learning Mode, and (2) a small panel that allows users to customize which emoji is displayed for each detected facial expression.

When in Observation Mode, users see real-time emotion badges and sarcasm markers placed unobtrusively in the corner of each participant's video tile. The goal is to enhance, rather than replace, natural facial cues. To support users who may interpret emojis differently, VideoVibes allows each user to choose which emoji corresponds to each of the seven expressions produced by the face recognition model. This design choice is motivated by research showing that autistic and neurotypical individuals sometimes assign different emotional meanings to the same emoji [13].

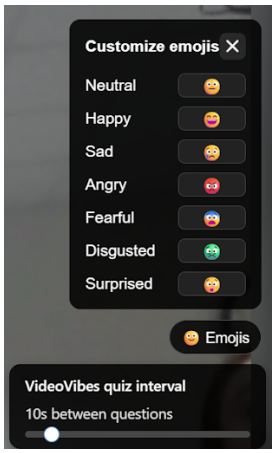

Fig. 1. Interface for selecting custom emojis for each detected expression.

Customizable emoji mapping helps ensure that system feedback remains intuitive and personally meaningful. The interface also includes a button to switch to Learning Mode, where real-time overlays are temporarily hidden so that users can practice identifying emotions based on live video. Because both modes use the same overall layout, switching between them is seamless.

C. Observation Mode

The system's default mode is Observation mode. The meeting functions as a normal Jitsi call, but the video tile of each participant includes a small emoji in the top-left corner that reflects their current facial expression. To keep the display stable, emotion predictions are averaged over a short window (typically two to five seconds) so the emoji does not change too quickly. Sarcasm cues appear separately. If someone says something that the system labels as sarcastic, a crossed-fingers emoji briefly appears in the bottom-right corner of their tile before disappearing. This gives a quick visual hint without cluttering the screen. Users can also adjust which emoji corresponds to each expression by opening the emoji selection panel. These changes apply immediately and allow users to choose the symbols that make the most sense to them.

Fig. 2. Example of a sarcasm indicator appearing during Observation Mode.

D. Learning Mode

Learning Mode is a toggleable feature designed to help users practice recognizing facial expressions on their own rather than relying entirely on automatic overlays. It can be turned on at any time using the graduation cap icon in the bottom toolbar.

When Learning Mode is active, the system pauses the normal emotion and sarcasm overlays and instead presents short quizzes. The user is shown a recent frame from another participant and asked to choose which emoji best matches the expression of that person. One option is correct and the other is a random alternative. A slider in the settings controls how often new questions appear.

After the user answers, a small feedback message appears: a confirmation if the choice was correct, or a brief explanation showing the correct emoji if it was not. These prompts are intended to give feedback without interrupting the flow of the meeting. Learning Mode can be turned off at any time by pressing the glasses icon, which returns the interface to regular Observation Mode.

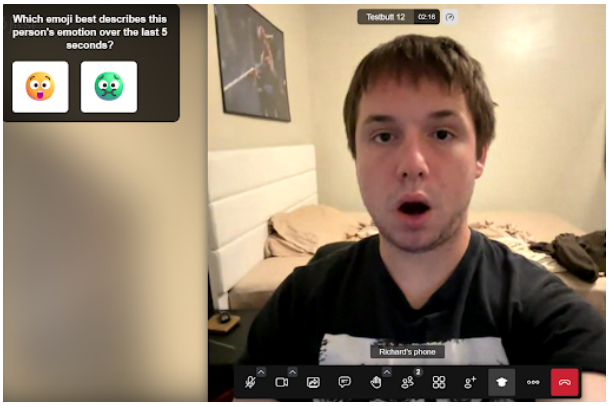

Fig. 3. Example of a quiz prompt shown during Learning Mode.

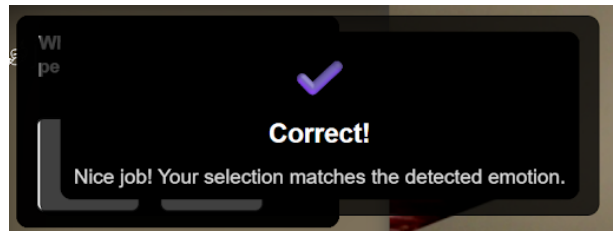

Fig. 4. Feedback shown after a correct response.

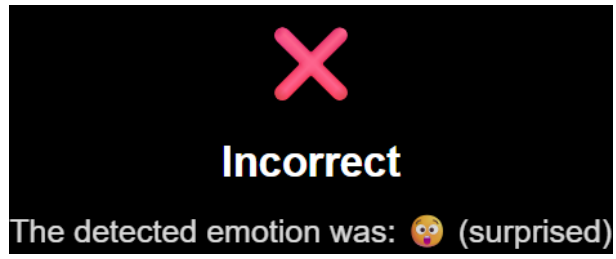

Fig. 5. Feedback shown after an incorrect response.

IV. User Evaluation

A key part of this project was to ensure that our product was able to successfully achieve our goal of increasing emotional fluency in a video call format. To this end, we conducted a user study to assess the effectiveness of our system using a combination of product walkthroughs and self-paced surveys.

A. Recruitment

Due to the relatively limited time frame for this project and the lack of easily available study recruitment methods, we used snowball sampling for this user study. In order to do this, each of the group members recruited participants via word of mouth and asked them to refer others whom they thought would be a good fit. Through this recruitment process, we got fourteen subjects, all college-aged. Although it would be ideal to have obtained a range of ages and to be able to specifically recruit participants with autism, it was not possible due to the recruitment resources we had available. This falls under future work for this project.

B. Study Design

For the interview phase of this project, we conducted interviews where the participant was logged onto a meeting by one interviewer and told that the other person in the meeting was another participant. However, this mock participant was actually another interviewer – this was done with the intent not to bias the real participant's responses, and so their behavior in the meeting would be as natural as possible.

These meetings were approximately five minutes, and the mock participant had the following requirements:

- Have a short conversation with the participant about any subject, while paying attention to go through at least 3 of the identifiable emotions
- Generate at least one sarcastic comment during the meeting

- When the participant is in learning mode, switch emotion every time they respond so that the next learning mode quiz is unique

Meanwhile, the interviewer was tasked with:

- Introducing the software and explaining the functionality of the program, specifically describing the difference between learning mode and observation mode
- Guiding the user to switch between the two modes and make sarcastic comments to test the functionality of the sarcasm detector
- Providing the participant with a Google Form for data collection, and remaining available for questions while the form was filled out

Through these meetings, we collected the data that is outlined in the following sections. The full text of the questionnaire can be found in Appendix **??**.

C. Ethics

To comply with the ethical standards set forth by the University, the data that was collected had no personally identifiable information (PII). In addition, the question asking participants about whether they had autism was optional, and this was reiterated in the instructions for the question as to not require any medical information that participants were not willing to give. All group members are also human research certified by the Texas A&M Institutional Review Board (IRB).

V. Results

Participants' responses to rate their ability to detect emotions in-person and over video calls averaged 4.423/5 and 3.429/5, respectively. Users who believed themselves to be a part of the target audience for VideoVibes saw an average increase of +1.3 in their rating of their ability to recognize the emotions of others over video calls. Those who did not see themselves as a part of the target audience rated themselves with an average of +0.364. When users were asked if they were less/more/equally capable of recognizing emotion over video calls after using VideoVibes, no user said they felt less capable, while 35.7% expressed more confidence, all others feeling equally capable. The same results held when asked if users felt less/more/equally capable of recognizing emotion in-person. However, even those who reported feeling 'equally' as capable rated their ability to identify emotions over video calls +0.44 on average after using VideoVibes. Those who recognized an increase in their confidence had an average increase in their ratings by +0.8. When asked to rate how well the emojis matched the expressions users in video calls had, users averaged a rating of 4.143/5. When asking users to freely give input on VideoVibes' emotional recognition, most users expressed that the emojis were receptive and fit the reality of how they were feeling well. Some participants felt burdened by technical shortcomings, like VideoVibes failing to pick up on certain expressions, being less responsive than they'd like, or occasionally misidentifying emotions. When asked to freely evaluate the sarcasm detection, all users found it to be both novel and effective, the biggest complaint being a high false positive rate (identifying normal speech as sarcastic).

TABLE I
Evaluation: Emotion Recognition

	Increase in Emotion Recognition over Video Call	**Perceived Increase in Emotion Recognition over Video Call**
Believe they are in the target audience	1.333	66.67%
Do not believe they are in the target audience	0.364	27.27%

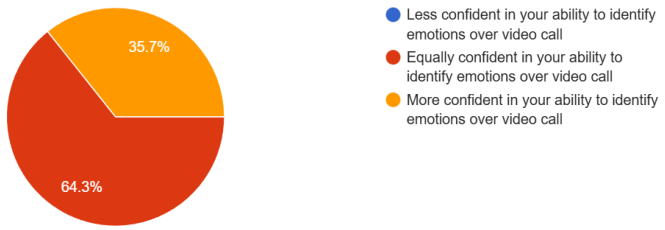

Fig. 6. Results concerning video calls when asked "After using the learning mode on VideoVibes, do you feel..."

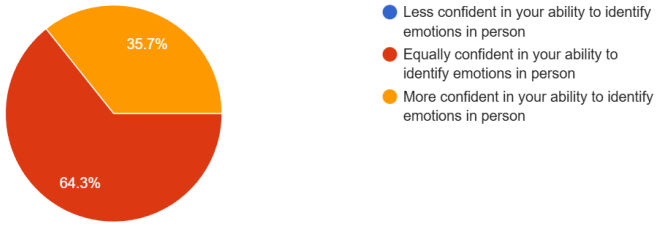

Fig. 7. Results concerning in person interaction when asked "After using the learning mode on VideoVibes, do you feel..."

VI. Discussion

Our initial assumptions based on our own intuition and what we saw in relevant literature was quickly confirmed seeing our users expressing their disparity in picking up on the emotions of others over video calls compared to in-person. Our target audience seeing a large increase (+1.333 out of 5) in their recognition of emotions gives us confidence that VideoVibes's solutions are appropriately tailored to assist those that we specifically sought to improve the user experience for. Given that those outside our target audience also noted a small improvement (and *none* said their experience was hindered i.e were annoyed), we can say that VideoVibes also has the elements of a universal design, improving the user experience in video calls for all who use it. Additionally, our critical feedback's focus on technological shortcomings indicate that the fundamentals to our approach to VideoVibes are solid, and

only hampered by the current sophistication of its implementation.

A. Limitations and Improvements

VideoVibes has several practical limitations, most of which come from building the system entirely with free tools and limited computing resources. Both the facial-expression model and the sarcasm classifier rely on lightweight, publicly available datasets and APIs. This keeps the system fast and easy to run, but it also reduces accuracy. The most common issue reported in our user study was false positives in sarcasm detection. Improving this would require models that incorporate tone of voice and richer training data, which are generally only available through paid APIs or large-scale datasets.

The system is also capped at around five simultaneous users. This limit was set to avoid excessive speech-to-text calls and to prevent the backend from being overloaded. With proper funding or a more scalable server, the same architecture could support larger meetings without changes to the core design.

Finally, our user study was small and relied mostly on college-aged, neurotypical participants recruited through word of mouth. We were not able to include autistic participants, even though they are a major target audience for the tool. A broader evaluation with a more representative group would be needed to fully assess VideoVibes' usefulness.

Overall, the limitations mostly reflect resource constraints rather than structural problems with the system. With stronger models, more compute, and a more diverse participant pool, VideoVibes could be expanded into a more accurate and widely usable accessibility tool.

VII. Conclusion and Future Work

VideoVibes provides real-time emotion and sarcasm cues during video calls by using lightweight facial-expression analysis and a simple audio-based sarcasm model. We provide this information in real time minimally obstructing the video calls. Our user study suggests that these cues can make online conversations easier to understand.

Future work could improve model accuracy, incorporate tone of voice, and bring the system to platforms like Zoom or Microsoft Teams. Furthermore we can add multi-moddal inputs to these models to add a shared context between the models. A broader user study, including autistic participants, would also give a clearer picture of how well the tool works. Overall, VideoVibes shows that small visual cues can make virtual communication more accessible and easier to follow.

References

[1] A. Persicke, J. Ranick, and M. St. Clair, "Teaching children with autism to detect and respond to sarcasm," *Research in Autism Spectrum Disorders*, vol. 7, no. 1, pp. 193–198, 2013.

[2] J. T. Hancock, M. Naaman, and K. Levy, "Ai-mediated communication: Definition, research agenda, and ethical considerations," *Journal of Computer-Mediated Communication*, vol. 25, no. 1, pp. 89–100, 2020.

[3] J. Daniels, J. N. Schwartz, C. Voss, N. Haber, A. Fazel, A. Kline, P. Washington, C. Feinstein, T. Winograd, and D. P. Wall, "Exploratory study examining the at-home feasibility of a wearable tool for social-affective learning in children with autism," *NPJ Digital Medicine*, vol. 1, p. 32, 2018.

[4] L. Dalle Nogare, A. Cerri, and A. M. Proverbio, "Emojis are comprehended better than facial expressions, by male participants," *Behavioral Sciences*, vol. 13, no. 3, p. 278, 2023.

[5] A. Hassouneh, A. M. Mutawa, and M. Murugappan, "Development of a real-time emotion recognition system using facial expressions and eeg based on machine learning and deep neural network methods," *Informatics in Medicine Unlocked*, vol. 20, p. 100372, 2020.

[6] W. Lin, "The application of real-time emotion recognition in video conferencing," *Journal of Computer Technology and Applied Mathematics*, vol. 1, no. 4, pp. 79–88, 2024.

[7] F. Diana *et al.*, "How video calls affect mimicry and trust during interactions," *Philosophical Transactions of the Royal Society B*, vol. 378, no. 1875, p. 20210484, 2023.

[8] N. Saffaryazdi, N. Kirkcaldy, G. Lee, K. Loveys, E. Broadbent, and M. Billinghurst, "Exploring the impact of computer-mediated emotional interactions on human facial and physiological responses," *Telematics and Informatics Reports*, vol. 14, p. 100131, 2024.

[9] J. Rollings *et al.*, "Facial behaviour and first impressions in computer mediated communication," *Computers in Human Behavior*, vol. 161, p. 108391, 2024.

[10] J. D. S. Ortega, P. Cardinal, and A. L. Koerich, "Emotion recognition using fusion of audio and video features," in *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, 2019.

[11] S.-J. Sun, A.-C. Huang, and W.-S. Ho, "Enhancing social skills in autism students with augmented-reality picturebooks," *Applied Sciences*, vol. 14, no. 11, p. 4907, 2024.

[12] K. Namikawa, I. Suzuki, R. Iijima, S. Sarcar, and Y. Ochiai, "Emojicam: Emoji-assisted video communication system leveraging facial expressions," in *Human-Computer Interaction: Design and User Experience Case Studies (HCII 2021)*, ser. Lecture Notes in Computer Science, vol. 12764. Springer, Cham, 2021, pp. 570–582.

[13] C. J. Hand, A. Kennedy, R. Filik, M. Pitchford, and C. M. Robus, "Emoji identification and emoji effects on sentence emotionality in asd-diagnosed adults and neurotypical controls," *Journal of Autism and Developmental Disorders*, vol. 53, no. 6, pp. 2514–2528, 2023.

[14] T. Kalandadze, C. F. Norbury, T. Nærland, and K.-A. B. Næss, "Figurative language comprehension in individuals with autism spectrum disorder: A meta-analytic review," *Autism*, vol. 22, no. 2, pp. 99–117, 2018.

[15] B. Mention *et al.*, "Humor in autism spectrum disorders: A systematic review," *L'Encéphale*, vol. 50, no. 2, pp. 200–210, 2024.

[16] F. G. E. Happé, "An advanced test of theory of mind: Understanding of story characters' thoughts and feelings by able autistic, mentally handicapped, and normal children and adults," *Journal of Autism and Developmental Disorders*, vol. 24, pp. 129–154, 1994.

[17] F. Panzeri, G. Mazzaggio, B. Giustolisi, S. Silleresi, and L. Surian, "The atypical pattern of irony comprehension in autistic children," *Applied Psycholinguistics*, vol. 43, no. 3, pp. 757–784, 2022.

[18] J. Kruger *et al.*, "Egocentrism over e-mail: Can we communicate as well as we think?" *Journal of Personality and Social Psychology*, vol. 89, no. 6, pp. 925–936, 2005.

[19] A. Kumar and G. Garg, "Empirical study of shallow and deep learning models for sarcasm detection using context in benchmark datasets," *Journal of Ambient Intelligence and Humanized Computing*, vol. 14, pp. 5327–5342, 2023.

[20] M. Kumar, A. Bansal, and R. Kaushal, "Spoken in jest, detected in earnest: A systematic review of sarcasm recognition—multimodal fusion, challenges, and future prospects," *Artificial Intelligence Review*, 2024.

[21] A. Joshi, V. Sharma, and P. Bhattacharyya, "Harnessing context incongruity for sarcasm detection," in *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Short Papers)*, Beijing, China, 2015, pp. 757–762.

[22] M. Franzen, "Developing emotion recognition for video conference software for autistic individuals," 2021. [Online]. Available: https://arxiv.org/abs/2101.10785

[23] N. Choque Olsson *et al.*, "Feasibility of internet-based multimodal emotion recognition training in adolescents with and without autism: A pilot study," *Internet Interventions*, vol. 41, p. 100861, 2025.

[24] J. C. Richardson, K. Swan, P. Lowenthal, and P. Ice, "Social presence in online learning: Past, present, and future," in *Proceedings of Global Learn 2016*. Association for the Advancement of Computing in Education, 2016,

pp. 477–483. [Online]. Available: https://patricklowenthal.com/social-presence-in-online-learning-past-present-and-future/
[25] Z. He, R. Du, and K. Perlin, "Lookatchat: Visualizing gaze awareness for remote small-group conversations," 2021.

Appendix

A: VideoVibes User Study

By filling out this form, you are consenting to your responses being used for a final project report for Human Computer Interaction. Please do not include any personally identifying information in any of your responses. Please do not fill out this form until you have finished the product demo with your interviewer.

How familiar are you with online video conferencing platforms (Zoom, Google Meet, etc)
1 (No familiarity) to 5 (Extremely familiar)

How well do you think you are able to accurately assess peoples' emotions IN PERSON?
1 (Not well at all) to 5 (Extremely well)

How well do you think you are able to accurately assess peoples' emotions OVER VIDEO CHAT?
1 (Not well at all) to 5 (Extremely well)

How well do you think you can identify sarcasm?
1 (Not well at all) to 5 (Extremely well)

When using VideoVibes, how well do you think you were able to accurately assess peoples' emotions?
1 (Not well at all) to 5 (Extremely well)

When using VideoVibes, how well do you think the emojis matched up with peoples' emotions?
1 (Not well at all) to 5 (Extremely well)

Do you have any additional comments on the emotion recognition portion of VideoVibes?

How do you feel about the sarcasm detection used in VideoVibes?

How was you experience using the learning mode for VideoVibes?

After using the learning mode on VideoVibes, do you feel…
Less confident in your ability to identify emotions over video call
Equally confident in your ability to identify emotions over video call
More confident in your ability to identify emotions over video call

After using the learning mode on VideoVibes, do you feel…
Less confident in your ability to identify emotions in person
Equally confident in your ability to identify emotions in person
More confident in your ability to identify emotions in person

Do you think you are part of the target audience for this product? (Mostly autistic individuals, but anybody who struggles with emotion or sarcasm detection, especially over video call)
Yes/No

Do you have or believe you have Autism Spectrum Disorder (ASD)? This is optional to answer.
Yes/No

If you have any additional comments about VideoVibes, please leave them here.